

1

Mark for Review



Line q in the xy -plane has a slope of $\frac{1}{7}$ and passes through the point $(0, 60)$. Which equation defines line q ?

A. $y = \frac{1}{7}x - 60$

B. $y = \frac{1}{7}x - 53$

C. $y = \frac{1}{7}x + 53$

D. $y = \frac{1}{7}x + 60$

2

Mark for Review



The function g is defined by $g(x) = -7x + 80$. What is the value of $g(x)$ when $x = -2$?

- A. $\frac{78}{7}$
- B. 66
- C. 71
- D. 94



Each face of a fair 12-sided die is labeled with a number from 1 through 12, with a different number appearing on each face. If the die is rolled one time, what is the probability of rolling a 2?

A. $\frac{1}{12}$

B. $\frac{2}{12}$

C. $\frac{10}{12}$

D. $\frac{11}{12}$

Student-produced response directions

- If you find **more than one correct answer**, enter only one answer.
- You can enter up to 5 characters for a **positive** answer and up to 6 characters (including the negative sign) for a **negative** answer.
- If your answer is a **fraction** that doesn't fit in the provided space, enter the decimal equivalent.
- If your answer is a **decimal** that doesn't fit in the provided space, enter it by truncating or rounding at the fourth digit.
- If your answer is a **mixed number** (such as $3\frac{1}{2}$), enter it as an improper fraction ($7/2$) or its decimal equivalent (3.5).
- Don't enter **symbols** such as a percent sign, comma, or dollar sign.

Examples

Answer	Acceptable ways to enter answer	Unacceptable: will NOT receive credit
3.5	3.5 3.50 7/2	$31/2$ 3 1/2

4



Mark for Review

$$6x + 28 = 6x + k$$

In the given equation, k is a constant. The equation has infinitely many solutions. What is the value of k ?

Answer Preview:



5

Mark for Review



Jasmin grows bean pods in her garden. This year, she harvested 490 bean pods and saved 10% of them to plant next year. How many of the harvested bean pods did Jasmin save to plant next year?

- A. 39
- B. 49
- C. 57
- D. 59

Student-produced response directions

- If you find **more than one correct answer**, enter only one answer.
- You can enter up to 5 characters for a **positive** answer and up to 6 characters (including the negative sign) for a **negative** answer.
- If your answer is a **fraction** that doesn't fit in the provided space, enter the decimal equivalent.
- If your answer is a **decimal** that doesn't fit in the provided space, enter it by truncating or rounding at the fourth digit.
- If your answer is a **mixed number** (such as $3\frac{1}{2}$), enter it as an improper fraction ($7/2$) or its decimal equivalent (3.5).
- Don't enter **symbols** such as a percent sign, comma, or dollar sign.

Examples

Answer	Acceptable ways to enter answer	Unacceptable: will NOT receive credit
3.5	3.5 3.50 7/2	$3\frac{1}{2}$ 3 1/2

6



Mark for Review

A triathlon is a multisport race consisting of three different consecutive events. A triathlon participant plans to complete the cycling event with an average speed of 19,008 yards per hour. What is this average speed, in miles per hour? (1 mile = 1,760 yards)

Answer Preview:

7



Mark for Review



Dan bought some apples and oranges. The apples cost \$0.79 per pound, and the oranges cost \$0.99 per pound. If Dan spent \$24.93 in total and bought twice as many pounds of oranges as pounds of apples, how many pounds of oranges did Dan buy?

(A) 9

(B) 9.5

(C) 18

(D) 22.16



Triangle R has an area of 80 square centimeters (cm^2). Square S has side lengths of 4 cm. What is the total area of triangle R and square S, in cm^2 ?

(A) 42

(B) 44

(C) 84

(D) 96



$$y = 3x + 4$$

$$y = 2x + 9$$

The solution to the given system of equations is (x, y) . What is the value of x ?

(A) 4

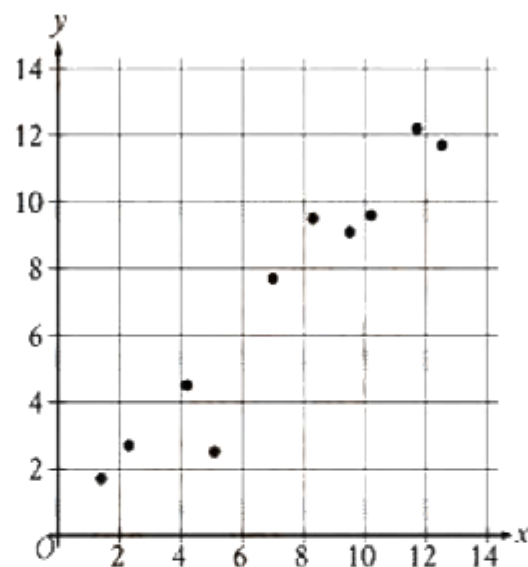
(B) 5

(C) 9

(D) 13



The scatterplot shows the relationship between two variables, x and y .



Which of the following equations is the most appropriate linear model for the data shown?

(A) $y = -2x$

(B) $y = -x$

(C) $y = x$

(D) $y = 2x$



A rectangle has a length that is 24 times its width. The function $y = (24w)(w)$ represents this situation, where y is the area, in square feet, of the rectangle and $y > 0$. Which of the following is the best interpretation of $24w$ in this context?

- (A) The length of the rectangle, in feet
- (B) The area of the rectangle, in square feet
- (C) The difference between the length and the width of the rectangle, in feet
- (D) The width of the rectangle, in feet



$$f(x) = 1,300(0.5)^{\frac{x}{14}}$$

The function f models the intensity of a beam, $f(x)$, in number of photons in the beam, x millimeters below the surface of a certain material. According to the model, what is the number of photons in the beam when it is at the surface of the material?

(A) 2

(B) 13

(C) 14

(D) 1,300



The gas mileage of a certain car is 22 miles per gallon when driven on the highway. If the car begins with 8 gallons of gas in the tank, which function predicts the amount of gas remaining in the tank, $f(d)$, after the car is driven d miles on the highway?

(A) $f(d) = \frac{d}{22} - 8$

(B) $f(d) = d - 8$

(C) $f(d) = 8 - \frac{d}{22}$

(D) $f(d) = 8 - d$



In triangle DEF , the measure of angle D is 46° and the measure of angle E is 104° . In triangle RST , the measure of angle R is 46° and the measure of angle S is 104° . Which of the following additional pieces of information is needed to determine whether triangle DEF is similar to triangle RST ?

- (A) The measure of angle F
- (B) The measure of angle T
- (C) The measure of angle F and the measure of angle T
- (D) No additional information is needed.

Student-produced response directions

- If you find **more than one correct answer**, enter only one answer.
- You can enter up to 5 characters for a **positive** answer and up to 6 characters (including the negative sign) for a **negative** answer.
- If your answer is a **fraction** that doesn't fit in the provided space, enter the decimal equivalent.
- If your answer is a **decimal** that doesn't fit in the provided space, enter it by truncating or rounding at the fourth digit.
- If your answer is a **mixed number** (such as $3\frac{1}{2}$), enter it as an improper fraction ($7/2$) or its decimal equivalent (3.5).
- Don't enter **symbols** such as a percent sign, comma, or dollar sign.

Examples

Answer	Acceptable ways to enter answer	Unacceptable: will NOT receive credit
3.5	3.5 3.50 7/2	$3\frac{1}{2}$ 3 1/2

15

Mark for Review

What is the slope of the graph of $y = \frac{1}{5}(19x + 19) + 7x$ in the xy -plane?

Answer Preview:



$$y = -4x - 2$$

One of the two equations in a system of linear equations is given. The system has no solution. Which of the following could be the second equation in this system?

(A) $y = \frac{1}{4}x - 2$

(B) $y = -4x - 8$

(C) $y = 4x + 8$

(D) $y = -\frac{1}{4}x + 2$

improper fraction ($7/2$) or its decimal equivalent (3.5).

- Don't enter **symbols** such as a percent sign, comma, or dollar sign.

Examples

Answer	Acceptable ways to enter answer	Unacceptable: will NOT receive credit
3.5	3.5 3.50 7/2	$31/2$ $3 \ 1/2$
$\frac{2}{3}$	$2/3$.6666 .6667 0.666 0.667	0.66 .66 0.67 .67
$-\frac{1}{3}$	-1/3 -.3333 -0.333	-.33 -0.33

17

Mark for Review

A linear model estimates the population of a city from 1992 to 2016. The model estimates the population was 61 thousand in 1992, 217 thousand in 2012, and x thousand in 2016. To the nearest whole number, what is the value of x ?

Answer Preview:

Student-produced response directions

- If you find **more than one correct answer**, enter only one answer.
- You can enter up to 5 characters for a **positive** answer and up to 6 characters (including the negative sign) for a **negative** answer.
- If your answer is a **fraction** that doesn't fit in the provided space, enter the decimal equivalent.
- If your answer is a **decimal** that doesn't fit in the provided space, enter it by truncating or rounding at the fourth digit.
- If your answer is a **mixed number** (such as $3\frac{1}{2}$), enter it as an improper fraction ($7/2$) or its decimal equivalent (3.5).
- Don't enter **symbols** such as a percent sign, comma, or dollar sign.

Examples

Answer	Acceptable ways to enter answer	Unacceptable: will NOT receive credit
3.5	3.5 3.50 7/2	$3\frac{1}{2}$ 3 1/2

18

Mark for Review

$$y = -1.5$$

$$y = x^2 + 4x + a$$

In the given system of equations, a is a positive integer constant. The system has no real solutions. What is the least possible value of a ?

Answer Preview:

Student-produced response directions

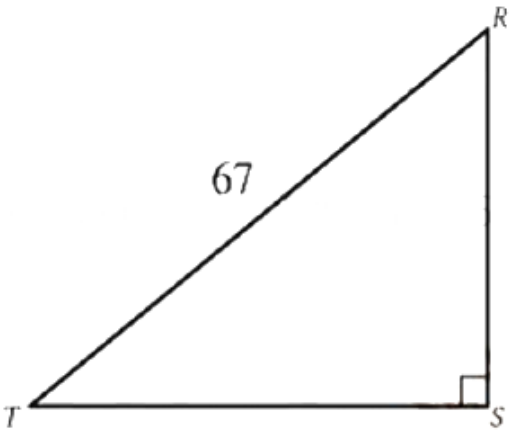
- If you find **more than one correct answer**, enter only one answer.
- You can enter up to 5 characters for a **positive** answer and up to 6 characters (including the negative sign) for a **negative** answer.
- If your answer is a **fraction** that doesn't fit in the provided space, enter the decimal equivalent.
- If your answer is a **decimal** that doesn't fit in the provided space, enter it by truncating or rounding at the fourth digit.
- If your answer is a **mixed number** (such as $3\frac{1}{2}$), enter it as an improper fraction ($7/2$) or its decimal equivalent (3.5).
- Don't enter **symbols** such as a percent sign, comma, or dollar sign.

Examples

Answer	Acceptable ways to enter answer	Unacceptable: will NOT receive credit
3.5	3.5	$3\frac{1}{2}$ 3 1/2
	3.50	
	$7/2$	
	$2/3$	

19

Mark for Review



Note: Figure not drawn to scale.

In the triangle shown, $RS = \sqrt{133}$. What is the value of $\sin R$?

Answer Preview:



$$(x + 13)^2 + 26(x + 13) + 169 = 0$$

How many distinct real solutions does the given equation have?

- ☐ (A) Exactly one
- ☐ (B) Exactly two
- ☐ (C) Infinitely many
- ☐ (D) Zero



$$(x + 3)^2 + (y - 19)^2 = 144$$

The graph of the given equation is a circle in the xy -plane. The point (a, b) lies on the circle. Which of the following is a possible value for a ?

(A) -16

(B) -13

(C) 12

(D) 19



$$f(x) = 8(7)^x$$

The function f is defined by the given equation. If $g(x) = f(x + 2)$, which of the following equations defines the function g ?

(A) $g(x) = 16(7)^x$

(B) $g(x) = 392(7)^x$

(C) $g(x) = 16(14)^x$

(D) $g(x) = 64(49)^x$